# Term Project — Part 2

Implementation, Deliverables and Grading

# Where We Are

**PART 1 ✓ Project Scope and Impairments (covered)**

- Scope, frame parameters, pilot pattern
- Multipath, Doppler grid, STO, CFO
- Adaptive Monte Carlo BER sweep

**PART 2 — Implementation, Deliverables and Grading (you are here)**

- Starter pack, TX / RX block diagrams
- What you build, allowed TX changes
- Deliverables, grading rubric, presentation Q&A

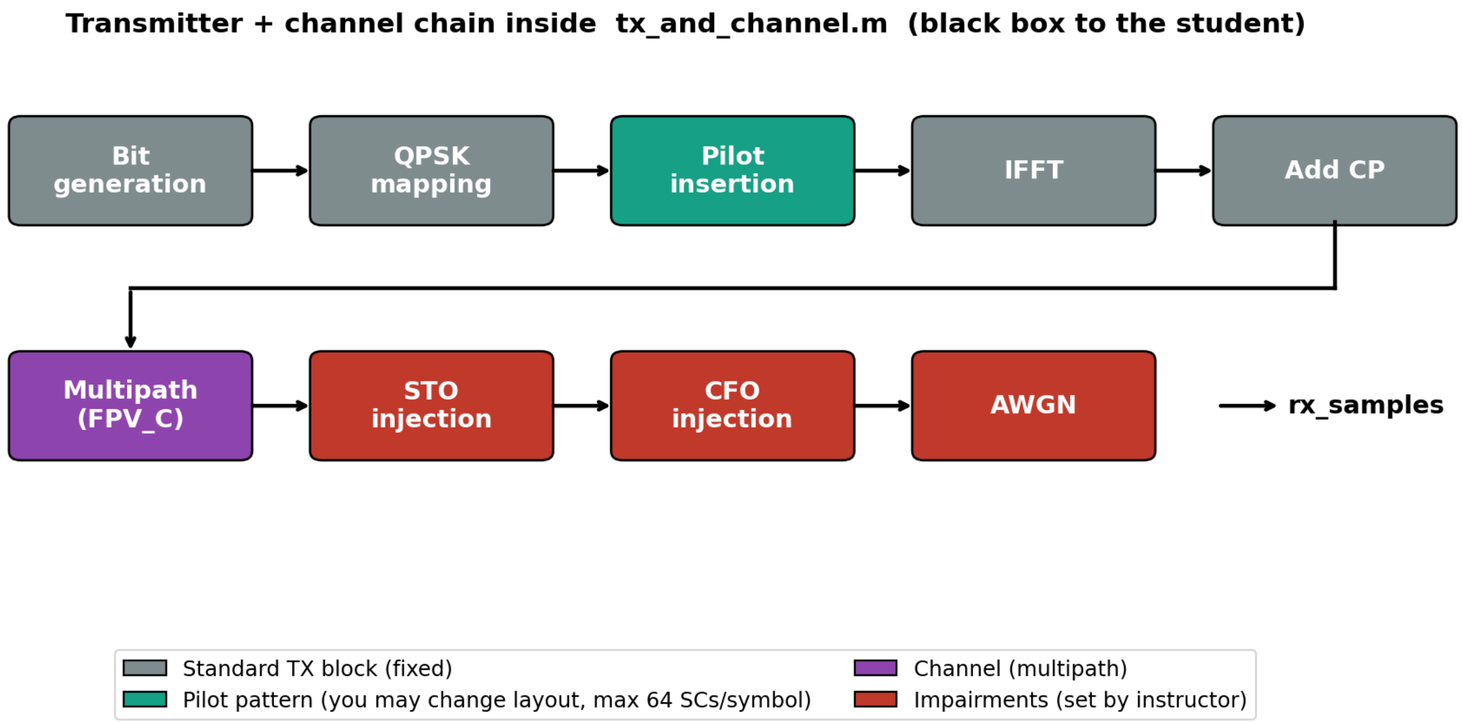
**You know the problem. Now: the tools, the deliverables, the grade.**

# The Starter Pack

| File | Role | You touch it? |
| --- | --- | --- |
| tx_and_channel.m | Black-box TX + channel + impairments | **no** |
| params_data.m | Frame parameter struct | **no** |
| ofdm_rx.m | Receiver orchestrator (calls sub-blocks) | **optional** |
| cp_remove.m | CP removal | **replaceable** |
| ofdm_demod.m | FFT / demodulation | **replaceable** |
| channel_estimate.m | LS-at-pilots + nearest-neighbour interp | **replaceable** |
| equalize.m | Zero-forcing one-tap equaliser | **replaceable** |
| qpsk_demap.m | Hard-decision QPSK demap | **replaceable** |
| sto_estimate_correct.m | Residual STO removal | **STUB — you implement** |
| cfo_estimate_correct.m | Residual CFO removal | **STUB — you implement** |
| doppler_track.m | Doppler-aware H[k] tracking | **STUB — you implement** |
| run_ber_sweep.m | Example BER vs Eb/N0 harness | **yours to adapt** |

**Three stubs to write. The other blocks are yours to swap if it helps.**

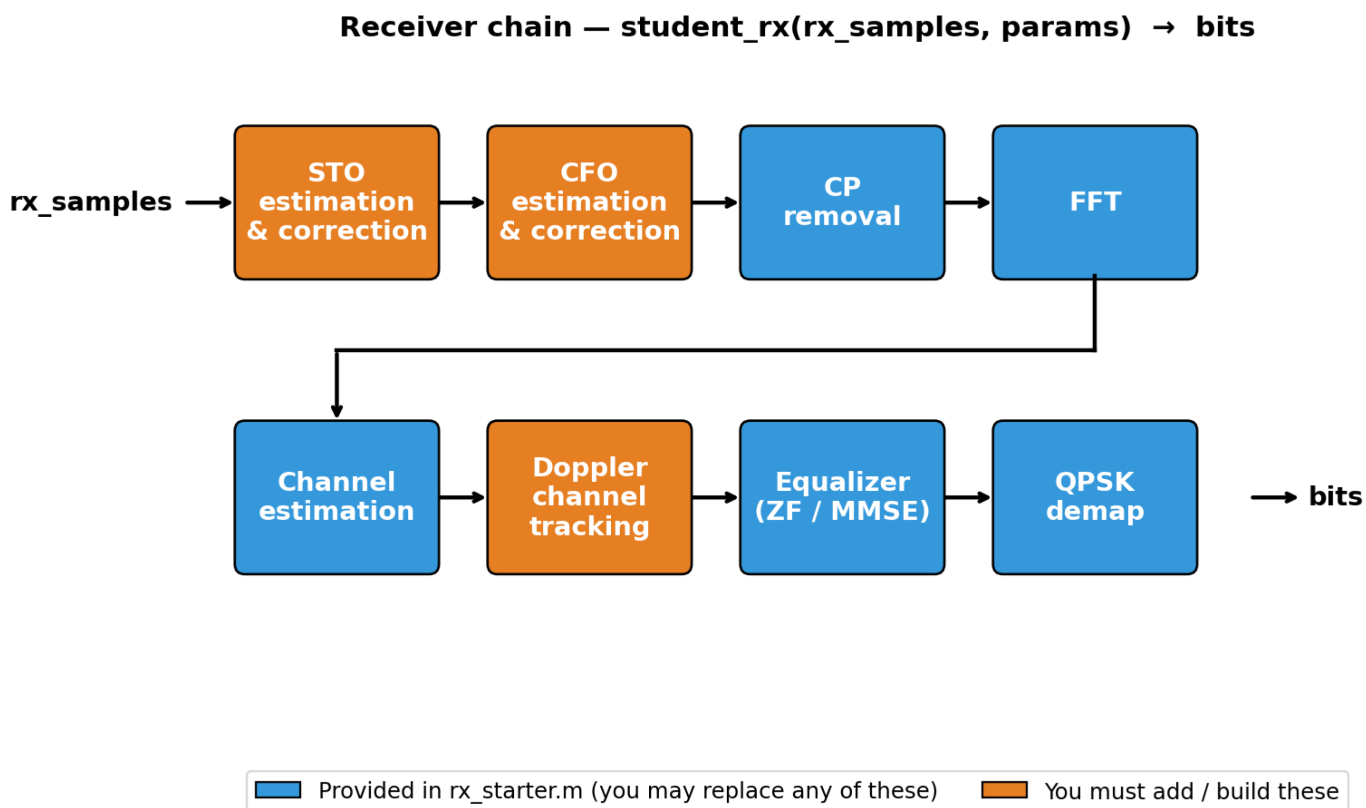
# TX Chain: The Black Box




**tx_and_channel.m runs all of this before you see a single sample.**

**This is fixed. You do not change anything left of "AWGN".**

# RX Chain — Your Job




`student_rx(rx_samples, params) → bits`

**Same skeleton as Lab 9. The interesting blocks are STO, CFO, channel tracking.**

# Where the Project Is Won

## 1. Residual STO

- Lab 9 phase-slope estimator (unwrap → polyfit), OR
- absorb it into the H[k] estimate.

Either is fine — pick the cheaper one.

## 2. Residual CFO

- M1 — pilot-aided estimator
- M2 — CP-based estimator
- M3 — decision-directed loop

Combine if it helps.

## 3. Doppler-aware tracking

- Starter: LS + nearest-neighbour.
- Upgrades: Wiener filter, decision-directed, Kalman, polynomial fit across symbols.

## 4. Optional: ICI-aware EQ

- 3-diagonal MMSE equaliser (method M4, Week 2).

- For top grades — trades op-count for BER at f_d = 3 kHz.

**Smarter channel tracking is usually where the biggest BER gain hides.**

# Allowed TX Modifications

## A. Interleaving

ALLOWED:
- Symbol-level interleaving
- Slot-level interleaving

Why:
- Spreads the impact of in-frame fades across the QPSK symbol stream.

NOT ALLOWED:
- Channel coding (FEC) of any kind.

Edit tx_and_channel.m to insert the matching interleaver — document it in the 1-page report.

## B. Pilot pattern

ALLOWED:
- Rearrange pilot positions (comb, scattered, diagonal, ...)

HARD CAP:
- $\leq$ 64 pilot SCs per OFDM symbol. (Raising this is OUT of scope.)

Why:
- Better pilot layout = better channel tracking at $f_d$ = 3 kHz.

Document the new pattern in the 1-page report.

**Two TX knobs are open — both must be justified in the report.**

# Three Files. One Email.

### 1. <lastname>_starter.zip

Your entire modified starter/ folder — zipped. Grader unzips, runs ofdm_rx(rx_samples, params).

### 2. <lastname>_report.pdf

1-page PDF. Methods you implemented and WHY · what worked / what didn't · MATLAB Profiler op-count.

### 3. <lastname>_ber.png

BER vs Eb/N0. One curve per $f_d \in \{0, 1000, 2000, 3000\}$ Hz (your RX) + Tier-1 baseline @ $f_d = 3000$ Hz on the same axes.

⚠ **File-naming is strict — wrong name = −5 points.**

**Three files, one email, one deadline. Don't lose 5 points to the filename.**

# Grading Rubric

| Component | Weight | How it is measured |
|---|---|---|
| BER performance | 50% | Sweep at f_d = 3000 Hz, held-out seed. Lower-Doppler curves required on PNG but do not score (they fuel the Q&A). |
| Complexity | 20% | MATLAB Profiler op-count on one frame. Lower → higher score. |
| Project presentation | 30% | Depth of understanding, defended under Q&A (in person or online). |

**TOTAL**

- Term-project total = 60% of course grade.
- With Lec 10/11 presentations (40%), this is the entire course — no separate exam.
- Absolute rubric, no quota — higher combined score = higher grade.

**Lower BER × lower op-count × clearer talk = higher grade.**

# Presentation — In Person or Online

| WHEN | WHERE | LENGTH |
|---|---|---|
| June 12, 2026 · 14:30 | in person · or · online | ~10 min talk + 5 min Q&A |

**Q1. Which impairment dominated your residual errors?**

Use your 4 Doppler curves to argue it.

**Q2. Why this STO / CFO / Doppler method —**

and what were the alternatives?

**Q3. If Δf jumped from 12 kHz to 20 kHz,**

which block breaks first?

**Q4. Explain one line of your code —**

physically, not syntactically.

*Internalise, don't memorise — same rubric in person or online.*

**In person or online — be ready to draw on a whiteboard and edit code on the spot.**

# Suggested Workflow

| # | Phase | What to do |
|---|---|---|
| 1 | Reproduce the baseline | Run run_ber_sweep.m as shipped — BER ≈ 40% at 28 dB confirms the chain runs. |
| 2 | Address residual STO | Look at the LS-estimate phase across SCs — fit the linear ramp (Lab 9) or absorb it. |
| 3 | Add residual CFO correction | Pick ONE M1–M3 method. Estimate $\Delta f$ from pilots or CP. Re-measure BER. |
| 4 | Add Doppler-aware tracking | Interpolate LS pilot estimates across symbols — linear first, spline/Wiener better. |
| 5 | Upgrade the channel estimator | LS + nearest-neighbour is a baseline, not a ceiling. Try Wiener / DD / Kalman. |
| 6 | (Optional) ICI-aware EQ | Replace ZF one-tap with 3-diagonal MMSE. Keep whichever wins on BER × op-count. |
| 7 | Write the 1-page report | Methods + why + Profiler op-count + final BER plot (all 4 Doppler curves). |
| 8 | Sanity-check the submission | Fresh MATLAB session, clean folder, end-to-end run. Fails in 5 min → grader fails. |

**Start with the baseline. Add one fix at a time and re-measure.**

# Rules and Final Word

## Hard rules

1. Individual work only.
   (High-level discussion OK,
   sharing code is not.)

2. Cite everything you borrow.
   Textbooks · papers · Week 2 talks.
   Failing to cite = academic dishonesty.

3. Reproducibility.
   No wall-clock seeding.

4. No 5G / LTE toolbox.
   Comm Toolbox only.

5. No internet / shell-outs.
   Pure MATLAB.

## Final word

"We don't grade the choice of method —
we grade your BER, your op-count, and
your ability to defend the design."

Pick something you understand deeply.
Implement it cleanly.
Be ready to explain it on the whiteboard
or over Zoom on June 12.

Three weeks.  One frame.  Own it.

Questions:
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